

Abhi Ramtel

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SUMMARY

Software Engineer with a strong background in backend systems and AI/ML, proficient in Python, TypeScript and modern AI frameworks such as LangChain. Experienced in building scalable platforms, high-performance data pipelines and integrating LLMs into production applications. Recognized for solving complex challenges under tight constraints and delivering reliable, high-throughput systems.

EXPERIENCE

Founding Engineer

April 2026 – Present

Stealth Fintech Startup

Chicago, IL

- Designed a payments platform (Auth, Ledger, Currency, Gateway) with fault-isolated service boundaries and horizontal scalability, enabling high-concurrency cross-border transactions at **99.9% uptime**.
- Engineered a secure digital wallet engine using **PostgreSQL row-level locking** and idempotency guarantees, eliminating double-spend risk and ensuring real-time balance accuracy under peak parallel load.
- Hardened platform security and compliance posture by implementing PII encryption, append-only audit logging, and a webhook event system, providing full transaction traceability and satisfying KYC/AML data requirements.
- Implemented CI/CD pipelines using GitHub Actions to automate build, test, and deployment processes, reducing release time by 40%.

Teaching Assistant, CSE 365: Computer Security

Aug 2025 – May 2026

University at Buffalo

Buffalo, NY

- Supported **90+ students** across weekly recitations and office hours for an upper-level security course, guiding hands-on SEED Labs covering buffer overflow exploitation, TCP attacks, packet sniffing/spoofing, Shellshock, and PKI.

Tech Consultant

Aug 2023 – May 2026

UB Information & Technology Department

Buffalo, NY

- Delivered Tier-2 technical support for **2,000+ users**, resolving complex OS, networking, and security incidents while managing enterprise imaging, configuration, and ticket-based workflows across Windows, macOS, and Linux environments.

Software Engineer Intern

Jan 2024 – May 2025

Cytocybernetics

Buffalo, NY

- Designed and shipped a next-generation Python data-pipeline feature built on vectorized execution, boosting throughput **60%** and enabling significantly more scalable data processing.
- Engineered a multi-threaded **C++/OpenMP** parallel processing engine for real-time analysis of over a **million** streaming electrophysiology datasets, achieving near-zero latency through optimized memory management and concurrent execution.
- Architected a new modular C/GTK framework with a GObject plug-in system and dynamic hardware-configuration modules, enabling extensible, maintainable, and reproducible scientific-simulation workflows.

PROJECTS

CarbonProxy | Python, FastAPI, SQLite, NumPy, Gemini API, GitHub Actions

Feb 2026

- Winner, Best Overall Hack – HackDuke 2026.** Architected semantic KV-caching middleware that reduced LLM session token consumption by **86%** (2,400→340 tokens) via compressed context state management.
- Designed REST API contracts and built an **async FastAPI backend** with LlamaIndex-powered cosine-similarity retrieval (0.75 threshold), enabling parallel development across 3 tracks for low-latency context injection

Dots | Swift, SwiftUI, ARKit, RoomPlan, RealityKit, Gemini API, Fetch.ai

Apr 2026

- 1st Place, Catalyst for Care; 3rd Place, UCLA Hacks Agentverse.** Built an AR accessibility assistant that improved indoor wayfinding for visually impaired users through spatial computing, voice interaction, agentic task automation, and AI-driven real-time navigation.

Adaptive Cloud RL Resource Manager | Python, PyTorch, NumPy, Gymnasium, CloudSim Plus

Aug – Dec 2025

- Implemented PPO, A2C, and DDQN (Reinforcement Learning Algorithms) from scratch to simulate adaptive cloud auto-scaling in a miniature data center, incorporating both **horizontal** and **vertical** scaling mechanisms.
- Demonstrated a **40% reduction in resource waste** without degrading task completion rates across diverse traffic scenarios.

ResQme | TypeScript, Node.js, C/C++, ESP32, Supabase

Oct 2025

- Winner, HackHarvard 2025.** Architected an **infrastructure-free ESP32 mesh network** with master election, encrypted payloads, and GPS telemetry for resilient SOS communication; built a REST admin API with real-time geolocation feeds and OTA update support.

TECHNICAL SKILLS

Languages: Python, Go, C/C++, Java, TypeScript/JavaScript, Rust, Swift, Shell Scripting

Backend & APIs: REST APIs, FastAPI, Microservices, gRPC, OAuth 2.0, JWT, Event-Driven Architecture, WebSockets, Flask, Django, Node.js

Databases: PostgreSQL, MySQL, SQLite, Firebase/Firestore, NoSQL – row-level locking, query optimization, schema design

Cloud & Infrastructure: AWS, GCP, Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins), Distributed Systems

AI & ML: PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn, LangChain, YOLOv8, OpenCV, LLMs, Semantic Embeddings, LlamaIndex

Tools and Methodologies: Git, JIRA, Agile/Scrum, Claude Code, Cursor

EDUCATION

University at Buffalo, The State University of New York

May 2026

B.S. Computer Science | Cybersecurity Minor | Cum Laude | Dean's List (2×)

Coursework: Data Structures & Algorithms, Distributed Systems, Reinforcement Learning, Machine Learning, Operating Systems, Computer Security, Networking Concepts, Software Engineering, Database Systems

Honors & Activities: 4× Hackathon Winner (HackHarvard, UCLA Hacks, HackIllinois, HackDuke), Nepali Student Association – Founding President, UB Lockdown NetDef – Red Team, Theta Tau – Professional Engineering Fraternity